

Rank Before You Merge: Native-Unit Selection for Visual Token Compression

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Abstract

Visual token compression depends on both token budget and selection representation. Rank-Before-Merge (RBM) scores complete native units from pre-merger feature norms without training or changing merger weights. At 25% retention, it exceeds Post-L2 on all nine text-dense model-benchmark pairs across Qwen3-VL, Qwen2.5-VL, and InternVL3, under model-specific taps and allocation rules. A full-split Qwen3-VL final-input control reduces the feature-depth confound: it gains 27.68 percentage points on TextVQA, 4.59 on DocVQA, and 235/1000 on OCRBench, but loses 5.64 points on GQA. Because Post-L2 aggregates main and auxiliary outputs, this is not a pure merger-order intervention. Rank, edge, swap, and FastV comparisons bound the interpretation: pre-merger magnitude is useful for the tested text-dense workloads, not a universal selector. Code: github.com/Code-Yan-ZX/qrbm-vlm.

Introduction

Modern VLMs produce long visual sequences from high-resolution images. Native spatial mergers convert neighboring patches into fewer language-model inputs; additional compression must choose which units survive. That choice depends on representation: a large norm before a learned projection need not remain large afterward.

RBM uses a deliberately simple, query-blind norm score to select complete pre-merger units while retaining native merger weights and interface dimensions. It adds neither training nor a learned compression module.

We separate operational accuracy from attribution. A three-model matrix compares deployed RBM with post-merger norm selection at the same final budget. A Qwen3-VL control instead scores the main merger’s final-block input with matched samples and token counts. This removes RBM’s early tap but not all implementation differences, because Post-L2 concatenates main and auxiliary outputs.

At 25% retention, RBM improves all nine text-dense Post-L2 comparisons; the final-input control preserves the direction on three text tasks, with a smaller DocVQA gain, while GQA reverses it. We contribute the interface-compatible rule, full-split operational and final-input evidence, and diagnostics that delimit its workload scope. Token retention denotes a task budget, not bit rate or total cost.

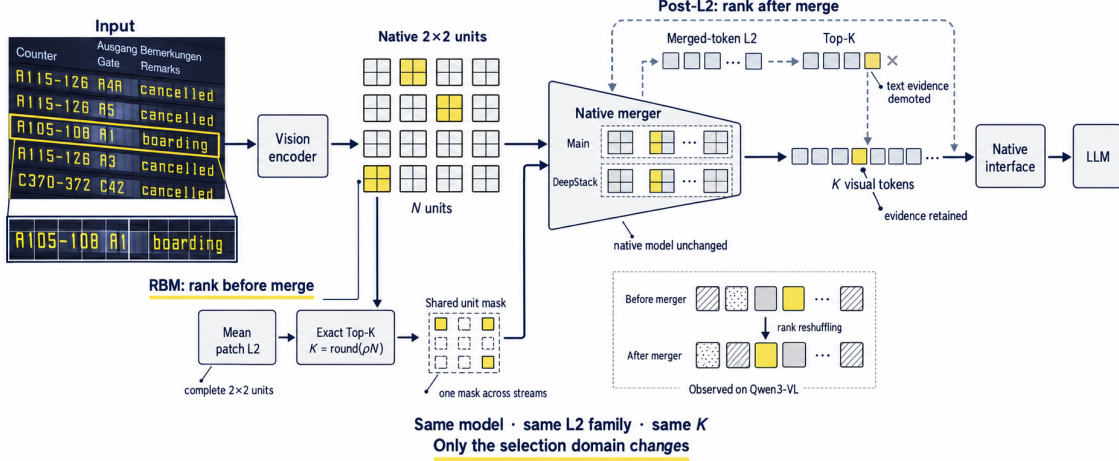


Figure 1: Native-unit selection schematic. Selected identities stay aligned with the surviving visual streams and interface. The illustrated pre-merger location is conceptual: operational Qwen3-VL RBM scores the layer-8 deepstack input, whereas the final-input control scores the main merger’s final-block input. The diagram’s ρ denotes retention, written κ in the text.

Related Work

Visual-token reduction can operate inside the language model through attention or progressive pruning [1, 2, 3, 4], or at the encoder output through [CLS] attention, graph propagation, fitted predictors, and multiple cues [5, 6, 7, 8, 9]. Unified systems combine saliency with diversity or text guidance [10, 11, 12]. Their signals and pruning stages differ from RBM’s image-only norm score, and equal final token count need not imply equal computation.

Merging changes the representation rather than only deleting tokens. ToMe, VisionZip, and AdaptMerge use matching, dominant-token preservation, and language guidance [13, 14, 15]; TokenPacker learns a compact projector, while GlimpsePrune, VScan, and AnchorPrune adapt reduction to inputs or model families [16, 17, 18, 19]. These methods motivate treating the selector’s input representation as part of the compression rule.

Recent selectors move reduction earlier or vary it across tokens and layers [20, 21, 22, 23, 24]; optimize coverage, reconstruction, or sensitivity [25, 26, 27, 28, 29]; and model semantics, information flow, or head importance [30, 31, 32, 33]. Layerwise, object-centric, local/global, and hybrid designs preserve evidence in other ways [34, 35, 36, 37, 38, 39]. Document pruning and provenance audits expose OCR-specific failures [40, 41], while LMMs-Eval and EffiVLM-Bench support broader evaluation [42, 43].

We focus on architectures with native spatial compression: Qwen2.5-VL, Qwen3-VL, and InternVL3 [44, 45, 46]. RBM retains their learned native operators and selects complete input units. Post-L2 is a diagnostic baseline from the same feature-magnitude family; FastV supplies a stronger attention-based comparison. The result-

ing evidence concerns these implementations and workloads, rather than a claim that downstream selectors generally assume rank preservation or that RBM ranks the full design space above.

Native-Unit Compression

Selection representation and token budget

Let an image contain N native spatial units, each grouping four patch vectors. At retention κ , retain

$$k = \max\{1, \text{round}(\kappa N)\}. \quad (1)$$

For features $h_{u,j}^{(t)}$ at tap t , RBM uses

$$a_u^{(t)} = \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}^{(t)}\|_2, \quad K_{\text{RBM}} = \text{TopK}_k\{a_u^{(t)}\}. \quad (2)$$

Post-L2 instead scores a returned visual vector z_u :

$$b_u = \|z_u\|_2, \quad K_{\text{post}} = \text{TopK}_k\{b_u\}. \quad (3)$$

Both rules use magnitude, but they evaluate different representations; they are not identical scoring functions applied to identical vectors. In Qwen3-VL, z_u concatenates the main and three deepstack outputs, whereas the other tested paths use their model-specific post-projector vectors.

In an idealized single stream at the merger input, $z_u = M(h_{u,1:4}^{(t)})$. For a fixed set, a per-unit merger gives identical retained outputs; rankings agree only when their top- k sets do. A strictly increasing $b_u = g(a_u)$ is sufficient at all budgets. We report Jaccard set overlap, but low overlap alone predicts neither utility nor accuracy. Qwen3-VL’s concatenated post vector departs from this idealization, preventing exact merger-only isolation.

Model-specific implementation

RBM retains complete units and aligns them with each adapter’s positions and auxiliary streams. Weights, interface dimensions, and decoding remain unchanged. Back-ends follow their compressed-position convention, which need not gather original spatial coordinates. Scoring costs $O(Nd)$ and heap top- k costs $O(N \log k)$; no learned parameters or LLM-attention pass are added.

Qwen3-VL RBM scores the layer-8 deepstack input and applies one set to four mergers; Post-L2 scores their concatenated return, changing feature depth, aggregation, and pruning location. Qwen2.5-VL scores the main-merger input, but its pre and post paths retain different ordering conventions. InternVL3 scores after pixel shuffle and before its projector; RBM budgets per tile while Post-L2 pools image tiles. These are operational comparisons, not isolated cross-model merger effects.

The Qwen3-VL *Pre-final* control scores final-block features at the main-merger input. Both arms execute every deepstack merger and match samples and per-image budgets. It removes the early tap, but still compares four input-patch norms with a concatenated main-plus-deepstack norm; it is neither a pure stage intervention nor an efficiency test.

Evaluation Protocol

We evaluate Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B in bf16 with greedy decoding. The main matrix and final-input control use vLLM 0.19 V1 [47]; FastV uses HuggingFace eager attention. Comparisons stay within backend, since small output checks do not establish full-split equivalence.

The full evaluation sets contain 5,000 TextVQA examples, 5,349 DocVQA examples, 1,000 OCRBench examples, and 12,578 GQA examples. Metrics are VQA accuracy, ANLS, OCRBench score out of 1,000, and normalized exact match, respectively [48, 49, 50, 51]. Scoring follows the corresponding lmms-eval rules [42]. The three former benchmarks are designated text-dense; GQA provides an object-centered question-answering contrast.

All comparisons retain 25% of native units and match per-image final counts. Qwen3-VL controls use native resolution, greedy decoding to 32 tokens, and vLLM eager execution; sequence limits are 8,192 with eight concurrent sequences, except DocVQA at 32,768 and four. Both compressed arms complete every sample; reused uncompressed anchors score two TextVQA and 18 OCRBench overruns as zero. Qwen2.5-VL OCRBench and InternVL3 DocVQA use four-million-pixel caps, precluding absolute cross-family ranking.

Final-input 95% intervals use 20,000 paired-bootstrap resamples of official per-example scores. Full-split scores and skipped-example handling are distinguished below.

Results

Operational accuracy across model families

Table 1 compares deployed RBM with Post-L2. RBM gains 11.0–38.4 percentage points on all nine text-dense VQA/ANLS cells or 298–432 OCRBench points, yet remains lossy versus no compression; Qwen3-VL DocVQA, for example, falls from .956 to .481.

GQA favors Post-L2 in all three point estimates. The small InternVL3 difference has an interval crossing zero in the paired analysis. The Qwen results give clearer evidence of a task boundary: pre-merger magnitude is useful for the tested text-dense workloads, but is not a general-purpose saliency criterion.

Final-input control

Table 2 separates the final-input contrast from early-tap RBM. Pre-final gains 27.68 points on TextVQA, 4.59 on DocVQA, and 235/1000 on OCRBench, but loses 5.64 on GQA. The smaller DocVQA gain than the operational 24.3 points prevents attributing that headline contrast to the merger boundary alone; residual score aggregation also limits merger-only attribution.

Comparison with attention-based selection

A same-harness full-OCRBench comparison evaluates RBM and FastV-k3 at a shared four-million-pixel cap and a 25% final budget (Table 3). Both methods use native

Table 1: Full-split operational results at 25% retention. None is uncompressed. Qwen3-VL RBM uses the early tap; this table does not isolate scoring stage. Deltas use unrounded scores and are percentage points except for OCRBench (/1000).

Model	Benchmark	None	RBM	Post-L2	Δ
Qwen3-VL	TextVQA	.844	.605	.222	+38.4
	DocVQA	.956	.481	.238	+24.3
	OCRBench	760	547	184	+363
	GQA	.616	.449	.477	-2.8
Qwen2.5-VL	TextVQA	.862	.702	.442	+26.1
	DocVQA	.949	.636	.526	+11.0
	OCRBench	817	480	182	+298
	GQA	.604	.559	.585	-2.6
InternVL3	TextVQA	.8338	.7890	.4148	+37.4
	DocVQA	.9221	.7284	.3820	+34.6
	OCRBench	852	753	321	+432
	GQA	.6293	.5993	.6031	-0.4

Table 2: Qwen3-VL final-input control, complete full splits. Pre-final scores four input patches; Post-L2 scores the concatenated main-plus-deepstack output. Deltas and 95% paired-bootstrap intervals are percentage points, except OCRBench points.

Benchmark	Pre-final	Post-L2	Δ [95% CI]
TextVQA	.4985	.2217	+27.68 [26.21, 29.19]
DocVQA	.2836	.2377	+4.59 [3.43, 5.78]
OCRBench	419	184	+235 [199, 271]
GQA	.4207	.4771	-5.64 [-6.42, -4.86]

positions, greedy decoding, and at most 32 generated tokens. The Qwen3-VL and Qwen2.5-VL runs encounter 79 and 88 common context or memory failures, respectively. These exact shared IDs receive zero in the scores out of 1,000; the completed paired sets contain 921 and 912 examples. RBM leads by 146 and 87 points. These HF scores are a separate experiment and must not be substituted into the vLLM matrix.

Diagnostics and Interpretation

The two operational Qwen3-VL rankings differ substantially. On diagnostic DocVQA, TextVQA, and GQA samples, Spearman correlations are .137, .332, and .360, and top-25% Jaccard overlaps are .180, .243, and .278. Representation choice therefore changes the selected set, although the different feature depths prevent merger-only attribution.

Rank shifts also associate with local edge energy: post-minus-pre rank correlations with Sobel energy are +.439, +.155, and +.036 in the same task order. On DocVQA, RBM-only versus Post-L2-only units have mean energy .641 versus .124, and 92%



Figure 2: Token-survival maps for audited TextVQA and DocVQA images. Each row shows operational early-tap RBM, Post-L2, and their overlap at the same 25% retention. Embedded scores and annotations belong to the historical $n = 200$ diagnostic; these selected cases illustrate spatial allocation rather than aggregate accuracy or a merger-only causal effect.

versus 35% exceed their image’s median. Because Sobel energy also responds to borders, texture, and contours, this establishes an edge association rather than text-specific or causal demotion. Figure 2 visualizes the spatial allocation in two audited cases.

A ranking-swap check fixes the post-merger computation but substitutes RBM’s kept identities. On the verified Qwen3-VL path, kept-set overlap is 1.000, retained representations are byte-identical, and predictions agree. This verifies path equivalence for a fixed set, not that norm or edge energy identifies causally relevant text. Qwen2.5-VL retains an ordering confound, while InternVL3 provides accuracy-level replication without byte-exact attribution.

Cost and Limitations

The retained-unit fraction k/N measures the final visual sequence budget, not an encoded bit rate or total inference cost. FastV processes all candidates through early language-model layers, while RBM does not eliminate vision-transformer work before

Table 3: Same-harness HF eager OCRBench comparison, 1,000 attempted examples per model. Common failures are scored zero; both methods retain the same final token count.

Model	RBM	FastV-k3	Δ (/1000)	Completed
Qwen3-VL	559	413	+146	921
Qwen2.5-VL	382	295	+87	912

its scoring tap; Pre-final deliberately executes all merger streams. Resolution, output length, batching, and backend also affect wall-clock performance.

On one A40, a Qwen3-VL TextVQA batch of 200 examples reaches 6.91 requests/s with RBM at 25% retention, versus 4.11 uncompressed and 6.89 with Post-L2. The vLLM run uses native resolution, at most eight concurrent sequences, greedy decoding capped at 32 tokens, and one untimed warmup; timing excludes engine construction and initial image loading. A separate five-run, one-example proxy averages .23 seconds for RBM and .36 seconds uncompressed. This supports an end-to-end gain for that workload, while the equal RBM/Post-L2 throughput shows no selector-specific speed advantage; the proxy is not a serving-latency distribution.

Our strongest controlled conclusion is the full-split Qwen3-VL final-input comparison, subject to its score-aggregation limitation. The three-model matrix extends the pattern across native merger designs without establishing a universal mechanism. Post-L2 is diagnostic, and FastV shows that RBM is not uniformly strongest. Shared loading failures also limit the HF OCRBench comparison. Broader selectors and direct text-region interventions remain necessary for general competitiveness or semantic claims.

Conclusion

Selecting native visual units from pre-merger features is a useful training-free compression design for the tested text-dense workloads. Separating the deployed RBM implementation from a final-input control shows that the useful signal persists at a later pre-merger tap, while score aggregation, feature depth, and stream handling also matter to operational performance. The GQA reversal and FastV comparisons define a clear boundary: selection representation should be evaluated against the intended workload, alongside the token budget and the model’s native interface.

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